How to make a poster, and how to pitch it

UMKC Quantum Computing Workshop
Saturday, October 24, 2026

Luke Miller, UMKC · QSAIC

1 Posters that won

2 Posters that did not

3 The checklist

4 Organizing and building yours

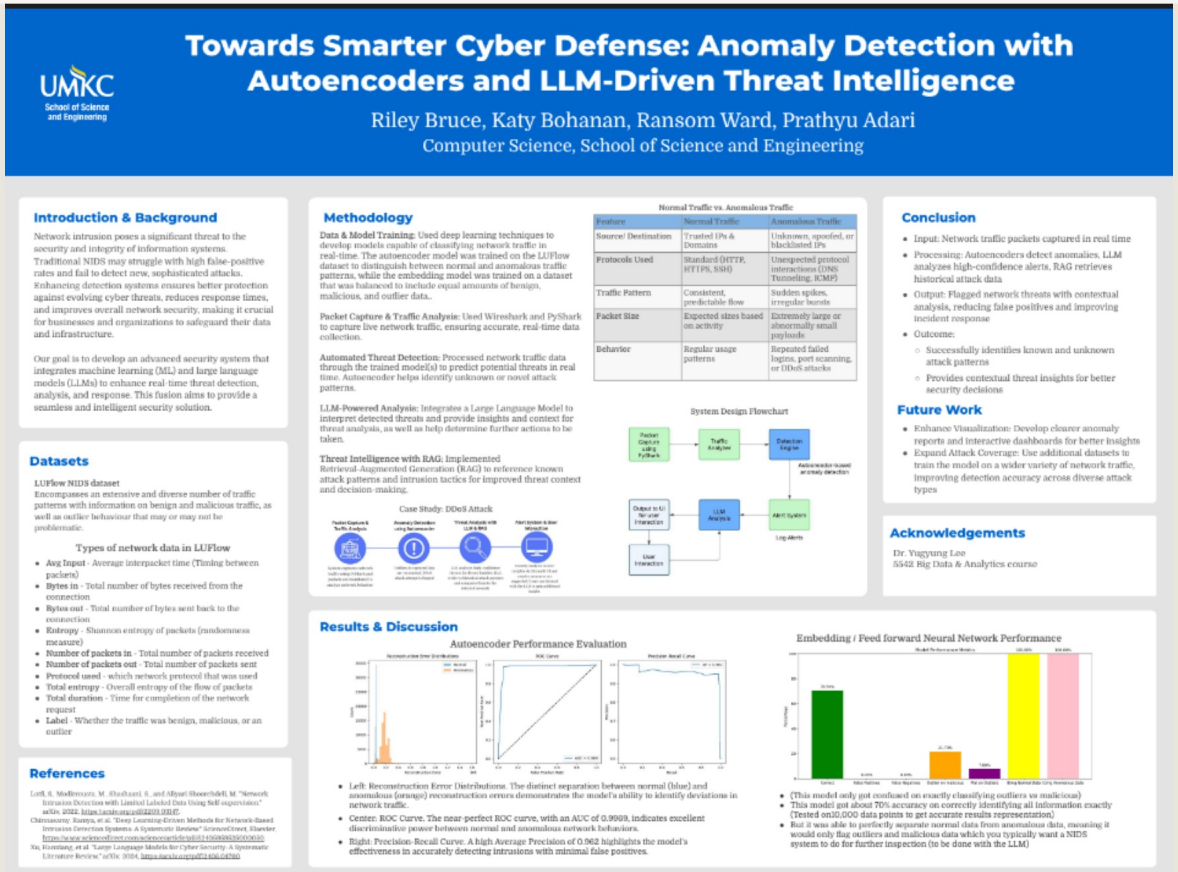
5 The elevator pitch

Look at posters, not at rules

- I could describe everything that makes a poster good. It is better to walk through real ones.
- These are from last spring's Hack-A-Roo.
- Decide for yourself before I say anything.

what we will look at	how many
first and second place posters	5
a runner up	1
posters that did not win	5
the checklist that falls out	1 slide

Winner: anomaly detection for cyber defense



Towards Smarter Cyber Defense: Anomaly Detection with Autoencoders and LLM-Driven Threat Intelligence

UMKC School of Science and Engineering

Riley Bruce, Katy Bohanan, Ransom Ward, Prathyu Adari

Computer Science, School of Science and Engineering

Introduction & Background

Network intrusion poses a significant threat to the security and integrity of information systems. Traditional NIDS may struggle with high false-positive rates and fail to detect new, sophisticated attacks. Enhancing detection systems ensures better protection against evolving cyber threats, reduces response times, and improves overall network security, making it crucial for businesses and organizations to safeguard their data and infrastructure.

Our goal is to develop an advanced security system that integrates machine learning (ML) and large language models (LLMs) to enhance real-time threat detection, analysis, and response. This fusion aims to provide a seamless and intelligent security solution.

Datasets

LUFlow NIDS dataset

Encompasses an extensive and diverse number of traffic patterns with information on benign and malicious traffic, as well as outlier behaviour that may or may not be problematic.

Types of network data in LUFlow

- **Avg Ipsut** - Average interpacket time (Timing between packets)
- **Bytes in** - Total number of bytes received from the connection
- **Bytes out** - Total number of bytes sent back to the connection
- **Entropy** - Shannon entropy of packets (randomness measure)
- **Number of packets in** - Total number of packets received
- **Number of packets out** - Total number of packets sent
- **Protocol used** - which network protocol that was used
- **Total entropy** - Overall entropy of the flow of packets
- **Total duration** - Time for completion of the network request
- **Label** - Whether the traffic was benign, malicious, or an outlier

References

Lotfi, S. Modiroozi, M. Khadivani, S. and Aliyari Shoorehdeli, M. "Network Intrusion Detection with Limited Labeled Data Using Self-supervision," arXiv, 2022. https://arxiv.org/pdf/2203.03147.

Chimiraarmy, Ramya, et al. "Deep Learning-Driven Methods for Network-Based Intrusion Detection Systems: A Systematic Review" ScienceDirect, Elsevier. https://www.sciencedirect.com/science/article/pii/S1546968325009436

Xu, Haoming, et al. "Large Language Models for Cyber Security: A Systematic Literature Review," arXiv, 2024. https://arxiv.org/pdf/2405.04760

Methodology

Data & Model Training: Used deep learning techniques to develop models capable of classifying network traffic in real-time. The autoencoder model was trained on the LUFlow dataset to distinguish between normal and anomalous traffic patterns, while the embedding model was trained on a dataset that was balanced to include equal amounts of benign, malicious, and outlier data.

Packet Capture & Traffic Analysis: Used Wireshark and PyShark to capture live network traffic, ensuring accurate, real-time data collection.

Automated Threat Detection: Processed network traffic data through the trained model(s) to predict potential threats in real time. Autoencoder helps identify unknown or novel attack patterns.

LLM-Powered Analysis: Integrates a Large Language Model to interpret detected threats and provide insights and context for threat analysis, as well as help determine further actions to be taken.

Threat Intelligence with RAG: Implemented Retrieval-Augmented Generation (RAG) to reference known attack patterns and intrusion tactics for improved threat context and decision-making.

Normal Traffic vs. Anomalous Traffic

Feature	Normal Traffic	Anomalous Traffic
Source/ Destination	Trusted IPs & Domains	Unknown, spoofed, or blacklisted IPs
Protocols Used	Standard (HTTP, HTTPS, SSH)	Unexpected protocol interactions (DNS Tunneling, ICMP)
Traffic Pattern	Consistent, predictable flow	Sudden spikes, irregular bursts
Packet Size	Expected sizes based on activity	Extremely large or abnormally small payloads
Behavior	Regular usage patterns	Repeated failed logins, port scanning, or DDoS attacks

System Design Flowchart

Case Study: DDoS Attack

Conclusion

- Input: Network traffic packets captured in real time
- Processing: Autoencoders detect anomalies, LLM analyzes high-confidence alerts, RAG retrieves historical attack data
- Output: Flagged network threats with contextual analysis, reducing false positives and improving incident response
- Outcome:
  - Successfully identifies known and unknown attack patterns
  - Provides contextual threat insights for better security decisions

Future Work

- Enhance Visualization: Develop clearer anomaly reports and interactive dashboards for better insights
- Expand Attack Coverage: Use additional datasets to train the model on a wider variety of network traffic, improving detection accuracy across diverse attack types

Acknowledgements

Dr. Yugyung Lee

5542 Big Data & Analytics course

Results & Discussion

Autoencoder Performance Evaluation

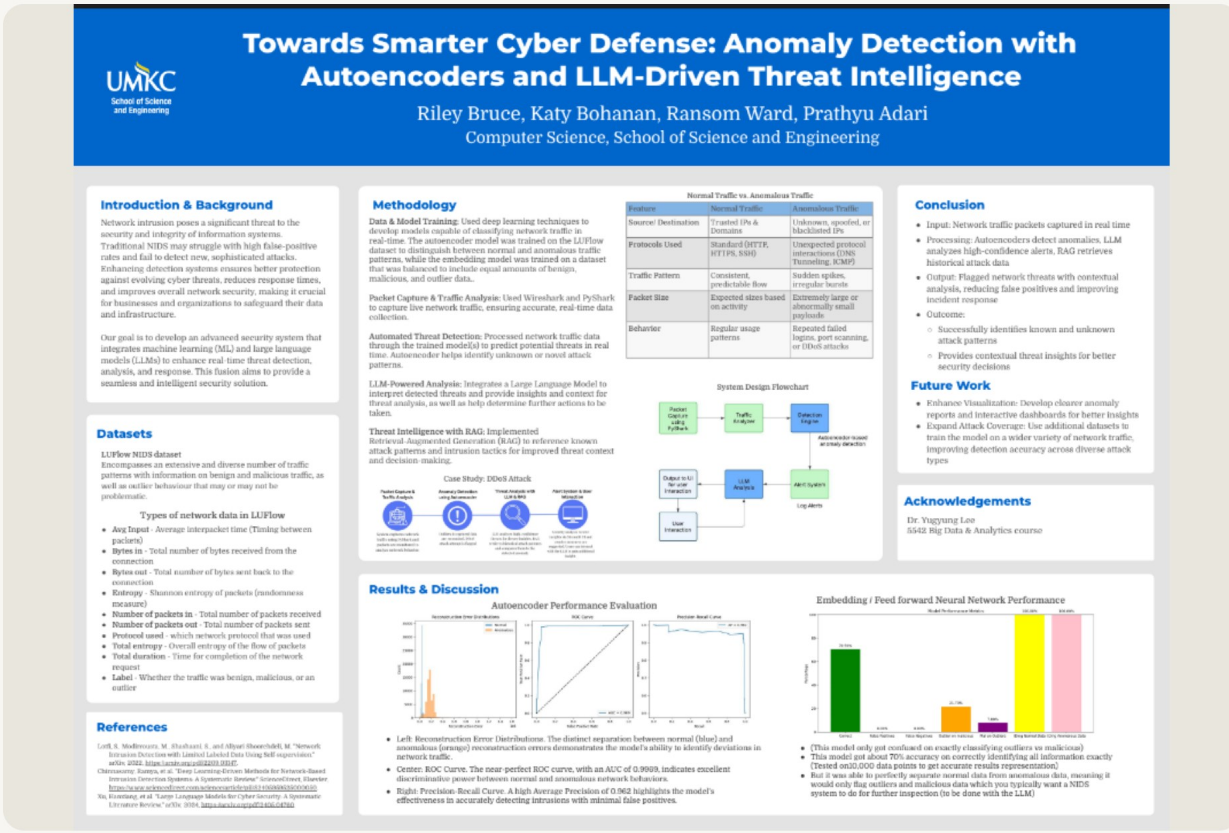
- Left: Reconstruction Error Distributions. The distinct separation between normal (blue) and anomalous (orange) reconstruction errors demonstrates the model's ability to identify deviations in network traffic.
- Center: ROC Curve. The near-perfect ROC curve, with an AUC of 0.9989, indicates excellent discriminative power between normal and anomalous network behaviors.
- Right: Precision-Recall Curve. A high Average Precision of 0.962 highlights the model's effectiveness in accurately detecting intrusions with minimal false positives.

Embedding / Feed forward Neural Network Performance

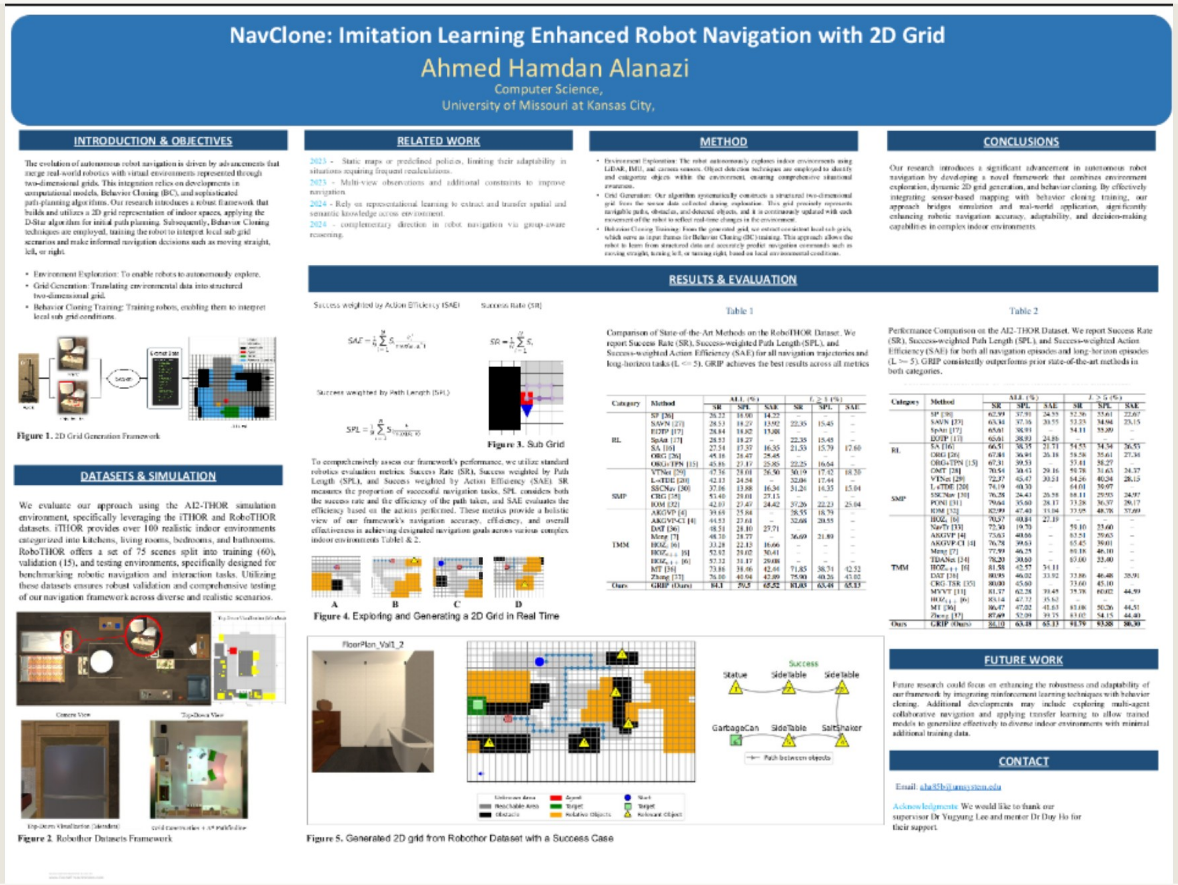
- (This model only got confused on exactly classifying outliers vs malicious)
- This model got about 70% accuracy on correctly identifying all information exactly (Tested on10,000 data points to get accurate results representation)
- But it was able to perfectly separate normal data from anomalous data, meaning it would only flag outliers and malicious data which you typically want a NIDS system to do for further inspection (to be done with the LLM)

Winner: anomaly detection for cyber defense

- The poster is not overcrowded.
- Colorful but tasteful graphics.
- Headings stand out from the text.
- Boxes break the poster into pieces.



Winner: robot navigation



NavClone: Imitation Learning Enhanced Robot Navigation with 2D Grid

Ahmed Hamdan Alanazi

Computer Science,
University of Missouri at Kansas City,

INTRODUCTION & OBJECTIVES

RELATED WORK

METHOD

CONCLUSIONS

RESULTS & EVALUATION

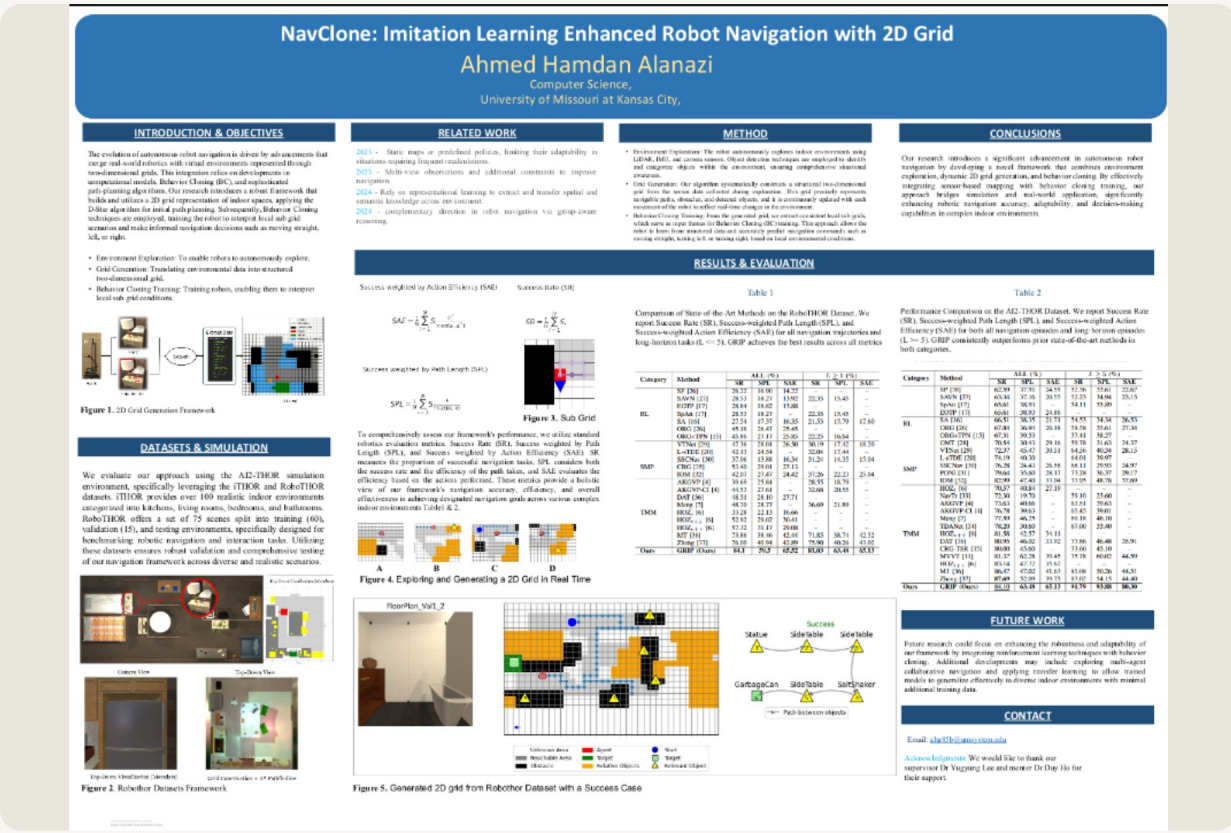
DATASETS & SIMULATION

FUTURE WORK

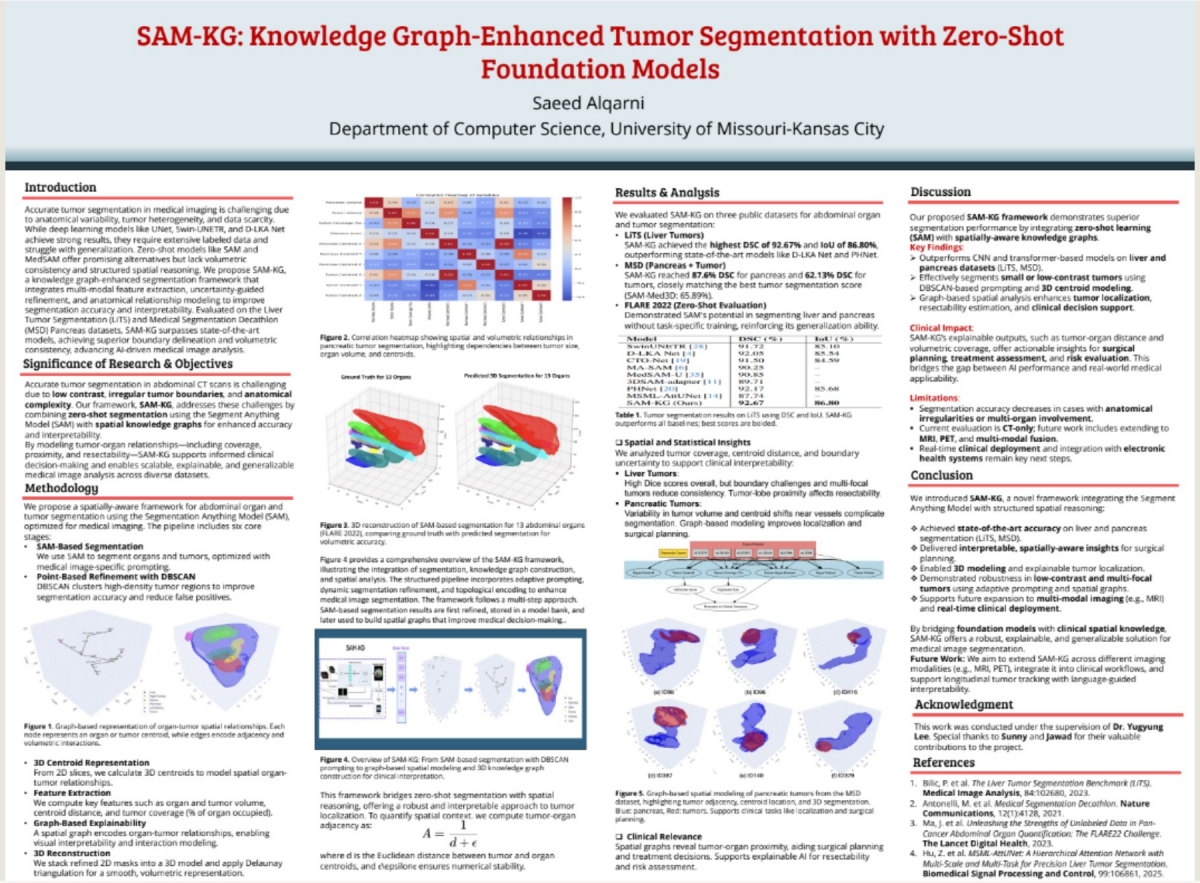
CONTACT

Winner: robot navigation

- They keep the text limited.
- The images let them tell a story.
- Header bars break the poster into elements.
- You can spot the results tables from far away.



Winner: tumor segmentation



SAM-KG: Knowledge Graph-Enhanced Tumor Segmentation with Zero-Shot Foundation Models

Saeed Alqarni
Department of Computer Science, University of Missouri-Kansas City

Introduction

Accurate tumor segmentation in medical imaging is challenging due to anatomical variability, tumor heterogeneity, and data scarcity. While deep learning models like UNet, Swin-UNETR, and D-LKA Net achieve strong results, they require extensive labeled data and struggle with generalization. Zero-shot models like SAM and MedSAM offer promising alternatives but lack volumetric consistency and structured spatial reasoning. We propose SAM-KG, a knowledge graph-enhanced segmentation framework that integrates multi-modal feature extraction, uncertainty-guided refinement, and anatomical relationship modeling to improve segmentation accuracy and interpretability. Evaluated on the Liver Tumor Segmentation (LiTS) and Medical Segmentation Decathlon (MSD) Pancreas datasets, SAM-KG surpasses state-of-the-art models, achieving superior boundary delineation and volumetric consistency, advancing AI-driven medical image analysis.

Significance of Research & Objectives

Accurate tumor segmentation in abdominal CT scans is challenging due to **low contrast**, **irregular tumor boundaries**, and **anatomical complexity**. Our framework, **SAM-KG**, addresses these challenges by combining **zero-shot segmentation** using the Segment Anything Model (SAM) with **spatial knowledge graphs** for enhanced accuracy and interpretability.
By modeling tumor-organ relationships—including coverage, proximity, and resectability—SAM-KG supports informed clinical decision-making and enables scalable, explainable, and generalizable medical image analysis across diverse datasets.

Methodology

We propose a spatially-aware framework for abdominal organ and tumor segmentation using the Segmentation Anything Model (SAM), optimized for medical imaging. The pipeline includes six core stages:

- **SAM-Based Segmentation**
  We use SAM to segment organs and tumors, optimized with medical image-specific prompting.
- **Point-Based Refinement with DBSCAN**
  DBSCAN clusters high-density tumor regions to improve segmentation accuracy and reduce false positives.

Figure 1. Graph-based representation of organ-tumor spatial relationships. Each node represents an organ or tumor centroid, while edges encode adjacency and volumetric interactions.

- **3D Centroid Representation**
  From 2D slices, we calculate 3D centroids to model spatial organ-tumor relationships.
- **Feature Extraction**
  We compute key features such as organ and tumor volume, centroid distance, and tumor coverage (% of organ occupied).
- **Graph-Based Explainability**
  A spatial graph encodes organ-tumor relationships, enabling visual interpretability and interaction modeling.
- **3D Reconstruction**
  We stack refined 2D masks into a 3D model and apply Delaunay triangulation for a smooth, volumetric representation.

Figure 2. Correlation heatmap showing spatial and volumetric relationships in pancreatic tumor segmentation, highlighting dependencies between tumor size, organ volume, and centroids.

Figure 3. 3D reconstruction of SAM-based segmentation for 13 abdominal organs (FLARE 2022), comparing ground truth with predicted segmentation for volumetric accuracy.

Figure 4 provides a comprehensive overview of the SAM-KG framework, illustrating the integration of segmentation, knowledge graph construction, and spatial analysis. The structured pipeline incorporates adaptive prompting, dynamic segmentation refinement, and topological encoding to enhance medical image segmentation. The framework follows a multi-step approach. SAM-based segmentation results are first refined, stored in a model bank, and later used to build spatial graphs that improve medical decision-making..

Figure 4. Overview of SAM-KG: From SAM-based segmentation with DBSCAN prompting to graph-based spatial modeling and 3D knowledge graph construction for clinical interpretation.

This framework bridges zero-shot segmentation with spatial reasoning, offering a robust and interpretable approach to tumor localization. To quantify spatial context, we compute tumor-organ adjacency as:

$$A = \frac{1}{d + \epsilon}$$

where d is the Euclidean distance between tumor and organ centroids, and epsilone ensures numerical stability.

Results & Analysis

We evaluated SAM-KG on three public datasets for abdominal organ and tumor segmentation:

- **LiTS (Liver Tumors)**
  SAM-KG achieved the **highest DSC of 92.67%** and **IoU of 86.80%**, outperforming state-of-the-art models like D-LKA Net and PHNet.
- **MSD (Pancreas + Tumor)**
  SAM-KG reached **87.6% DSC** for pancreas and **62.13% DSC** for tumors, closely matching the best tumor segmentation score (SAM-Med3D: 65.89%).
- **FLARE 2022 (Zero-Shot Evaluation)**
  Demonstrated SAM's potential in segmenting liver and pancreas without task-specific training, reinforcing its generalization ability.

Model	DSC (%)	IoU (%)
SwinUNETR [28]	91.72	85.10
D-LKA Net [1]	92.05	85.84
CTFD-Net [39]	91.50	84.59
MA-SAM [61]	90.25	–
MedSAM-U [35]	90.85	–
3DSAM-adapter [11]	89.71	–
PHNet [20]	92.17	85.68
MSML-AttUNet [14]	87.74	–
SAM-KG (Ours)	**92.67**	**86.80**

Table 1. Tumor segmentation results on LiTS using DSC and IoU. SAM-KG outperforms all baselines; best scores are bolded.

❑ **Spatial and Statistical Insights**
We analyzed tumor coverage, centroid distance, and boundary uncertainty to support clinical interpretability:

- **Liver Tumors**:
  High Dice scores overall, but boundary challenges and multi-focal tumors reduce consistency. Tumor-lobe proximity affects resectability.
- **Pancreatic Tumors**:
  Variability in tumor volume and centroid shifts near vessels complicate segmentation. Graph-based modeling improves localization and surgical planning.

Figure 5. Graph-based spatial modeling of pancreatic tumors from the MSD dataset, highlighting tumor adjacency, centroid location, and 3D segmentation. Blue: pancreas, Red: tumors. Supports clinical tasks like localization and surgical planning.

❑ **Clinical Relevance**
Spatial graphs reveal tumor-organ proximity, aiding surgical planning and treatment decisions. Supports explainable AI for resectability and risk assessment.

Discussion

Our proposed **SAM-KG framework** demonstrates superior segmentation performance by integrating **zero-shot learning (SAM)** with **spatially-aware knowledge graphs**.
Key Findings:

- Outperforms CNN and transformer-based models on **liver and pancreas datasets** (LiTS, MSD).
- Effectively segments **small or low-contrast tumors** using DBSCAN-based prompting and **3D centroid modeling**.
- Graph-based spatial analysis enhances **tumor localization**, resectability estimation, and **clinical decision support**.

Clinical Impact:
SAM-KG's explainable outputs, such as tumor-organ distance and volumetric coverage, offer actionable insights for **surgical planning**, **treatment assessment**, and **risk evaluation**. This bridges the gap between AI performance and real-world medical applicability.

Limitations:

- Segmentation accuracy decreases in cases with **anatomical irregularities or multi-organ involvement.**
- Current evaluation is **CT-only**; future work includes extending to **MRI**, **PET**, and **multi-modal fusion.**
- Real-time **clinical deployment** and integration with **electronic health systems** remain key next steps.

Conclusion

We introduced **SAM-KG**, a novel framework integrating the Segment Anything Model with structured spatial reasoning:

- Achieved **state-of-the-art accuracy** on liver and pancreas segmentation (LiTS, MSD).
- Delivered **interpretable, spatially-aware insights** for surgical planning.
- Enabled **3D modeling** and explainable tumor localization.
- Demonstrated robustness in **low-contrast and multi-focal tumors** using adaptive prompting and spatial graphs.
- Supports future expansion to **multi-modal imaging** (e.g., MRI) and **real-time clinical deployment**.

By bridging **foundation models** with **clinical spatial knowledge**, SAM-KG offers a robust, explainable, and generalizable solution for medical image segmentation.
Future Work: We aim to extend SAM-KG across different imaging modalities (e.g., MRI, PET), integrate it into clinical workflows, and support longitudinal tumor tracking with language-guided interpretability.

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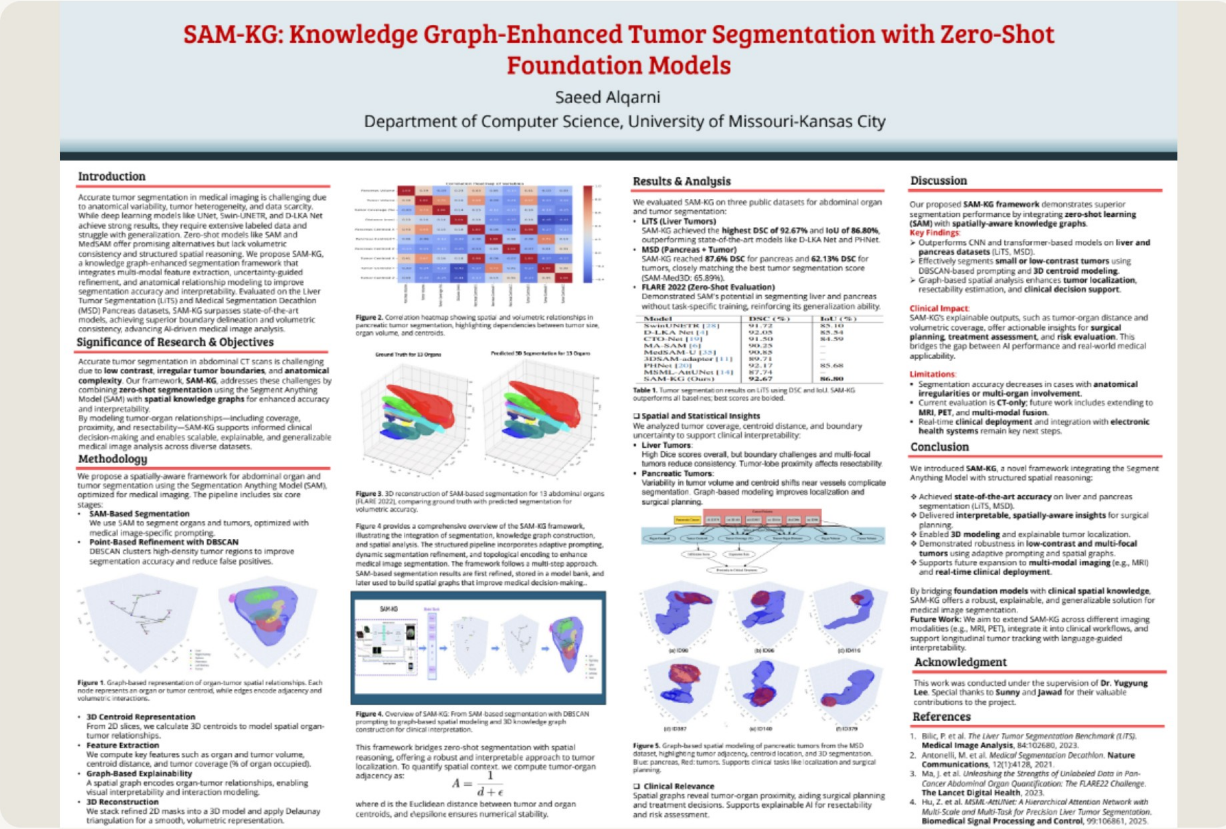
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References

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Winner: tumor segmentation

- A little denser, but graphics break up the text boxes.
- Consistent color.
- The graphics are interesting.
- Title and headings stand out, as before.



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Saeed Alqarni

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Winner: AI-powered travel stories



UMKC School of Science and Engineering

WanderTales: AI-Powered Travel Stories & Recommendations

Experience Your Journey Before It Begins – Smart AI Travel Planning

Viswanth Tamanna, Dharani Thakkallapally, Pavan Sundar Reddy, Banu Teja Jampani

Background

- Existing travel recommendation systems provide generic, static suggestions without considering real-time updates, user preferences, or contextual relevance, making trip planning inefficient and impersonal.
- This project leverages retrieval-augmented generation (RAG) and FAISS-based vector search to dynamically curate travel recommendations based on user interests, recent trends, and real-time data.
- Unlike conventional planners, this system not only assists in travel planning but also generates AI-driven travel narratives, offering users a seamless way to document and relive their experiences.

Introduction & Objective

This project integrates APIs like **OpenAI, Amadeus API, Google Maps API, Weather API, SERP API, and Geoapify API** to offer real-time, dynamic travel recommendations tailored to user preferences.

Objectives:

- Provides users with an interactive platform to customize their trips via a chatbot, ensuring recommendations align with their interests and real-time travel conditions.
- Generates engaging travel stories and video previews, allowing users to visualize their journey and build anticipation before the trip.

Methodology

- Travelers enter their destination, dates, and preferences, which serve as the foundation for recommendations.
- **API-Based Data Retrieval:**
  - **Amadeus API** fetches flight details and itineraries.
  - **OpenWeather API (SerpAPI)** provides real-time weather updates.
  - **Wikipedia & SerpAPI** gather historical and cultural information.
  - **GPT-4 (LLM)** generates personalized travel insights.
- A **Travel Recommendation System** suggests itineraries, while the **RAG Engine** refines results by retrieving relevant travel information.
- The **AI Travel Story Teller** generates engaging narratives using GPT-4, complemented by DALL·E for images and additional tools for video storytelling.
- Users receive **text-based stories, AI-generated travel images, and immersive video experiences** with background audio.
- A web-based **front-end interface** allows users to access recommendations, explore AI-generated stories, and customize their itinerary.

Comparison of Methods

DALL·E	Stable diffusion
Captures details well	May interpret vaguely
Illustrative, slightly artificial, vibrant	More photorealistic but sometimes muted
Strong, well-structured, proportionate	Can struggle with fine details, distortions possible
Concept art, fantasy, storytelling	Realistic landscapes, textures, artistic freedom

Analysis

Results & Discussion

Offers personalized itineraries, Integrates video and AI-generated stories to enhance user engagement. system ensures flexibility and convenience, adapting to users' changing preferences for a seamless travel experience.

Example: Trip from **Kansas City** to **Manali** (March 21 - March 31, 2025). Summary of travel plan generated

- Fly from Kansas City to Delhi, then take a local flight to Manali. Upon arrival, check in to the hotel for a relaxed start.
- Visit **Hidimba Temple, Manali Sanctuary, Solang Valley**, and experience the breathtaking **Rohtang Pass**. Dine at charming cafes like **Johnson's Café, Il Forno, Fat Plate Café, and Lazy Dog** while enjoying local cuisine.
- Fly back to Delhi and explore iconic attractions like **Qutub Minar, Humayun's Tomb, and India Gate**. Visit bustling markets like **Chandni Chowk**, try local delicacies at recommended restaurants, and immerse yourself in Delhi's vibrant street culture.

Conclusion & Future Scope

- Our AI-powered travel companion transforms trip planning into an immersive adventure, delivering personalized itineraries, stunning visuals, and interactive previews. With real-time insights and AI-driven storytelling, we make every journey exciting before it even begins.
- Future scope includes Expanding to AR/VR travel simulations, multilingual AI storytelling, interactive voice guides, real-time data feeds, booking platform integration, and personalized AI-generated travel videos featuring user integration.

References

Zhang, X., Zhao, Y., & Li, X. (2017). *Personalized Travel Recommendation System: A Survey*. International Journal of Information Technology & Decision Making, 16(3), 579-583.

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Brown, T., & Smith, A. (2019). *AI-Based Storytelling for Personalized Content Generation*. Proceedings of the International Conference on Natural Language Processing, 34-45.

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Sample video output

Departure : Kansas
Destination: Manali
Start date: 21-03-2025
End date: 31-03-2025
Purpose of visit: Romantic trip

Acknowledgments

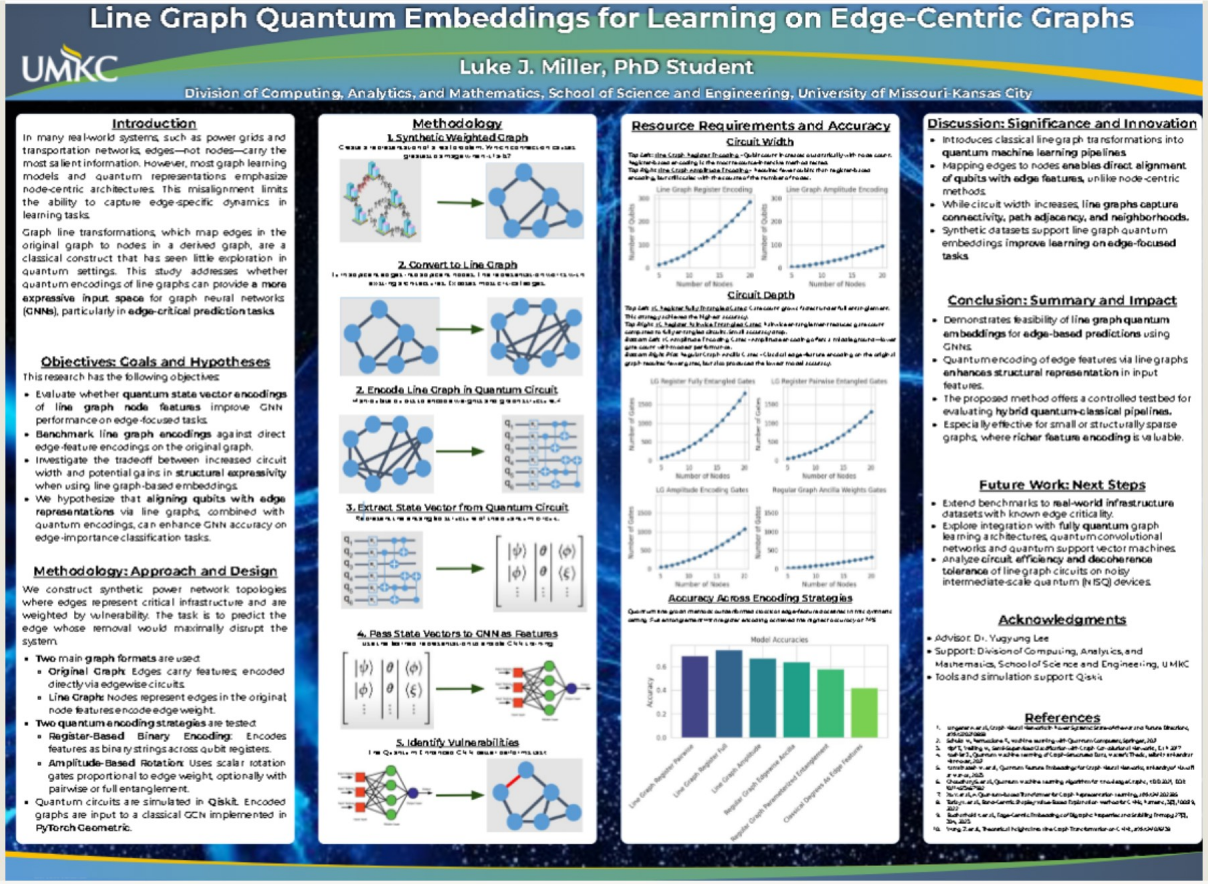
We sincerely thank Dr. Yugyung Lee for teaching us the latest technologies like RAG, Lora and encouraging us to apply them in our project. The CS 5588 - Data Science Capstone course provided us with the opportunity to explore AI-driven travel storytelling. We also appreciate the open-source community and API providers like Amadeus, OpenWeather, Wikipedia, and OpenAI for their resources.

Winner: AI-powered travel stories

- I am not sure what is going on with that wave, but I want to find out.
- Limited text.
- Clear headings.
- If it were mine, I would want more white space between the columns.

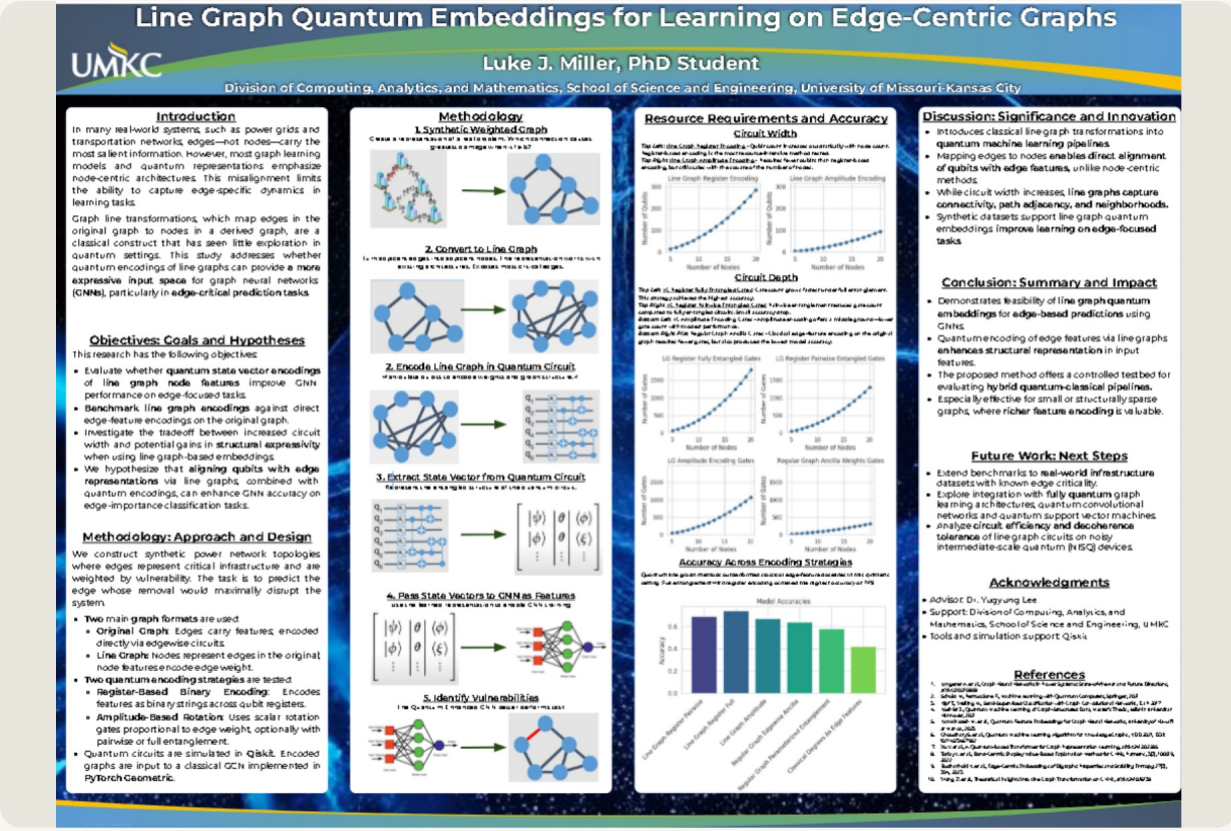


Winner: line graph quantum embeddings

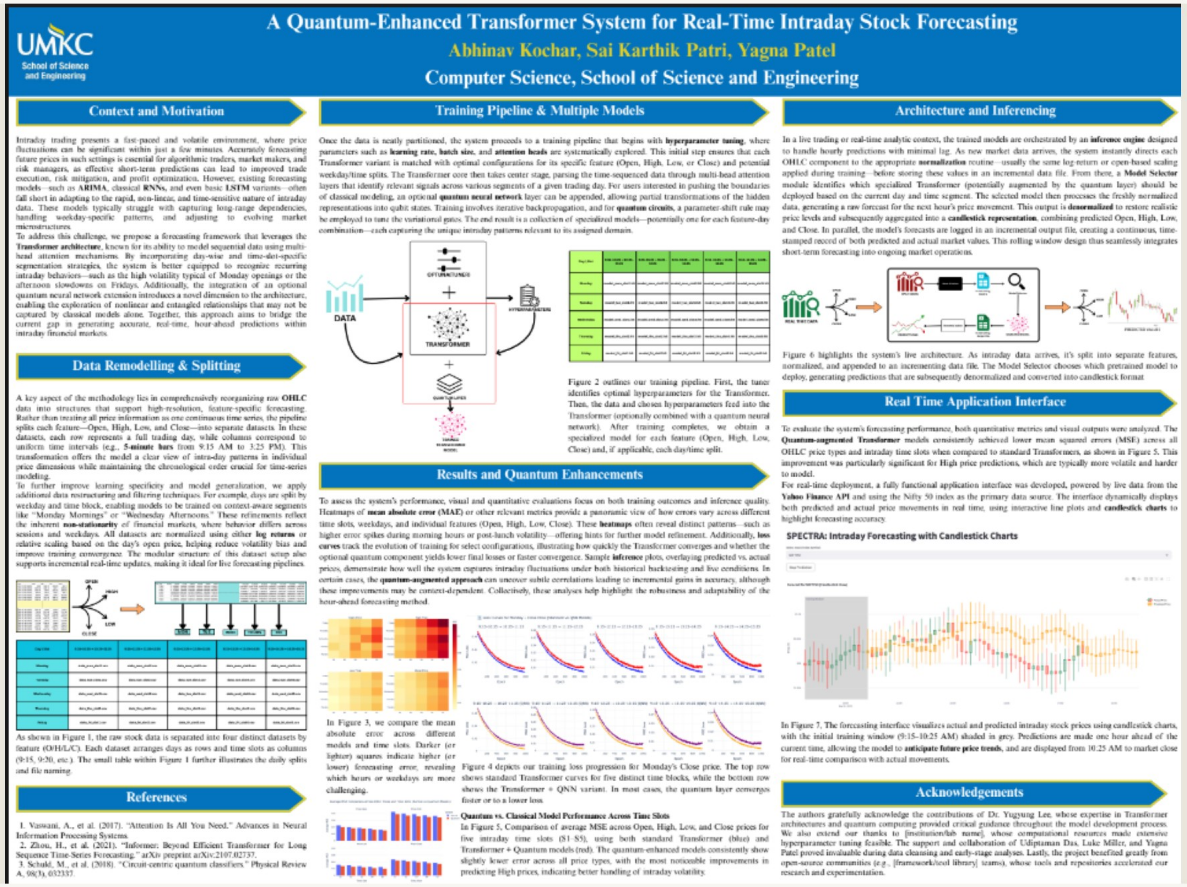


Winner: line graph quantum embeddings

- Instead of color, it uses space to mark the headings.
- The graphics are central.
- Boxes break up the poster.



Runner up: quantum-enhanced stock forecasting



A Quantum-Enhanced Transformer System for Real-Time Intraday Stock Forecasting

Abhinav Kochar, Sai Karthik Patri, Yagna Patel
Computer Science, School of Science and Engineering
UMKC School of Science and Engineering

Context and Motivation

Intraday trading presents a fast-paced and volatile environment, where price fluctuations can be significant within just a few minutes. Accurately forecasting future prices in such settings is essential for algorithmic traders, market makers, and risk managers, as effective short-term predictions can lead to improved trade execution, risk mitigation, and profit optimization. However, existing forecasting models—such as **ARIMA**, classical **RNNs**, and even basic **LSTM** variants—often fall short in adapting to the rapid, non-linear, and time-sensitive nature of intraday data. These models typically struggle with capturing long-range dependencies, handling weekday-specific patterns, and adjusting to evolving market microstructures.

To address this challenge, we propose a forecasting framework that leverages the **Transformer architecture**, known for its ability to model sequential data using multi-head attention mechanisms. By incorporating day-wise and time-slot-specific segmentation strategies, the system is better equipped to recognize recurring intraday behaviors—such as the high volatility typical of Monday openings or the afternoon slowdowns on Fridays. Additionally, the integration of an optional quantum neural network extension introduces a novel dimension to the architecture, enabling the exploration of nonlinear and entangled relationships that may not be captured by classical models alone. Together, this approach aims to bridge the current gap in generating accurate, real-time, hour-ahead predictions within intraday financial markets.

Data Remodelling & Splitting

A key aspect of the methodology lies in comprehensively reorganizing raw **OHLC** data into structures that support high-resolution, feature-specific forecasting. Rather than treating all price information as one continuous time series, the pipeline splits each feature—Open, High, Low, and Close—into separate datasets. In these datasets, each row represents a full trading day, while columns correspond to uniform time intervals (e.g., **5-minute bars** from 9:15 AM to 3:25 PM). This transformation offers the model a clear view of intra-day patterns in individual price dimensions while maintaining the chronological order crucial for time-series modeling.

To further improve learning specificity and model generalization, we apply additional data restructuring and filtering techniques. For example, days are split by weekday and time block, enabling models to be trained on context-aware segments like "Monday Mornings" or "Wednesday Afternoons." These refinements reflect the inherent **non-stationarity** of financial markets, where behavior differs across sessions and weekdays. All datasets are normalized using either **log returns** or relative scaling based on the day's open price, helping reduce volatility bias and improve training convergence. The modular structure of this dataset setup also supports incremental real-time updates, making it ideal for live forecasting pipelines.

As shown in Figure 1, the raw stock data is separated into four distinct datasets by feature (OHLC). Each dataset arranges days as rows and time slots as columns (9:15, 9:30, etc.). The small table within Figure 1 further illustrates the daily splits and file naming.

References

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Training Pipeline & Multiple Models

Once the data is neatly partitioned, the system proceeds to a training pipeline that begins with **hyperparameter tuning**, where parameters such as **learning rate, batch size,** and **attention heads** are systematically explored. This initial step ensures that each Transformer variant is matched with optimal configurations for its specific feature (Open, High, Low, or Close) and potential weekday/time splits. The Transformer core then takes center stage, parsing the time-sequenced data through multi-head attention layers that identify relevant signals across various segments of a given trading day. For users interested in pushing the boundaries of classical modeling, an optional **quantum neural network** layer can be appended, allowing partial transformations of the hidden representations into qubit states. Training involves iterative backpropagation, and for **quantum circuits**, a parameter-shift rule may be employed to tune the variational gates. The end result is a collection of specialized models—potentially one for each feature-day combination—each capturing the unique intraday patterns relevant to its assigned domain.

Figure 2 outlines our training pipeline. First, the tuner identifies optimal hyperparameters for the Transformer. Then, the data and chosen hyperparameters feed into the Transformer (optionally combined with a quantum neural network). After training completes, we obtain a specialized model for each feature (Open, High, Low, Close) and, if applicable, each day/time split.

Results and Quantum Enhancements

To assess the system's performance, visual and quantitative evaluations focus on both training outcomes and inference quality. Heatmaps of **mean absolute error (MAE)** or other relevant metrics provide a panoramic view of how errors vary across different time slots, weekdays, and individual features (Open, High, Low, Close). These **heatmaps** often reveal distinct patterns—such as higher error spikes during morning hours or post-lunch volatility—offering hints for further model refinement. Additionally, **loss curves** track the evolution of training for select configurations, illustrating how quickly the Transformer converges and whether the optional quantum component yields lower final losses or faster convergence. Sample **inference** plots, overlaying predicted vs. actual prices, demonstrate how well the system captures intraday fluctuations under both historical backtesting and live conditions. In certain cases, the **quantum-augmented approach** can uncover subtle correlations leading to incremental gains in accuracy, although these improvements may be context-dependent. Collectively, these analyses help highlight the robustness and adaptability of the hour-ahead forecasting method.

In Figure 3, we compare the mean absolute error across different models and time slots. Darker (or lighter) squares indicate higher (or lower) forecasting error, revealing which hours or weekdays are more challenging.

Figure 4 depicts our training loss progression for Monday's Close price. The top row shows standard Transformer curves for five distinct time blocks, while the bottom row shows the Transformer + QNN variant. In most cases, the quantum layer converges faster or to a lower loss.

Quantum vs. Classical Model Performance Across Time Slots

In Figure 5, Comparison of average MSE across Open, High, Low, and Close prices for five intraday time slots (S1–S5), using both standard Transformer (blue) and Transformer + Quantum models (red). The quantum-enhanced models consistently show slightly lower error across all price types, with the most noticeable improvements in predicting High prices, indicating better handling of intraday volatility.

Architecture and Inferencing

In a live trading or real-time analytic context, the trained models are orchestrated by an **inference engine** designed to handle hourly predictions with minimal lag. As new market data arrives, the system instantly directs each OHLC component to the appropriate **normalization** routine—usually the same log-return or open-based scaling applied during training—before storing these values in an incremental data file. From there, a **Model Selector** module identifies which specialized Transformer (potentially augmented by the quantum layer) should be deployed based on the current day and time segment. The selected model then processes the freshly normalized data, generating a raw forecast for the next hour's price movement. This output is **denormalized** to restore realistic price levels and subsequently aggregated into a **candlestick representation**, combining predicted Open, High, Low, and Close. In parallel, the model's forecasts are logged in an incremental output file, creating a continuous, time-stamped record of both predicted and actual market values. This rolling window design thus seamlessly integrates short-term forecasting into ongoing market operations.

Figure 6 highlights the system's live architecture. As intraday data arrives, it's split into separate features, normalized, and appended to an incrementing data file. The Model Selector chooses which pretrained model to deploy, generating predictions that are subsequently denormalized and converted into candlestick format.

Real Time Application Interface

To evaluate the system's forecasting performance, both quantitative metrics and visual outputs were analyzed. The **Quantum-augmented Transformer** models consistently achieved lower mean squared errors (MSE) across all OHLC price types and intraday time slots when compared to standard Transformers, as shown in Figure 5. This improvement was particularly significant for High price predictions, which are typically more volatile and harder to model.

For real-time deployment, a fully functional application interface was developed, powered by live data from the **Yahoo Finance** API and using the Nifty 50 index as the primary data source. The interface dynamically displays both predicted and actual price movements in real time, using interactive line plots and **candlestick charts** to highlight forecasting accuracy.

SPECTRA: Intraday Forecasting with Candlestick Charts

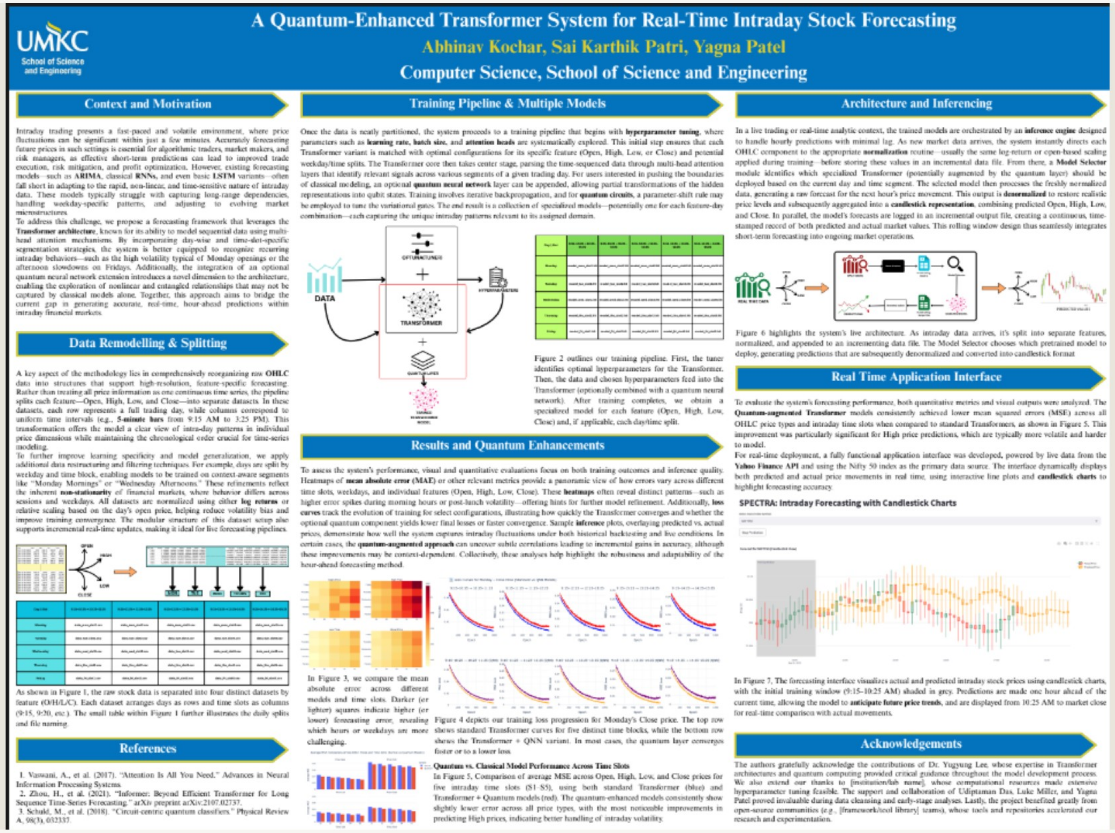
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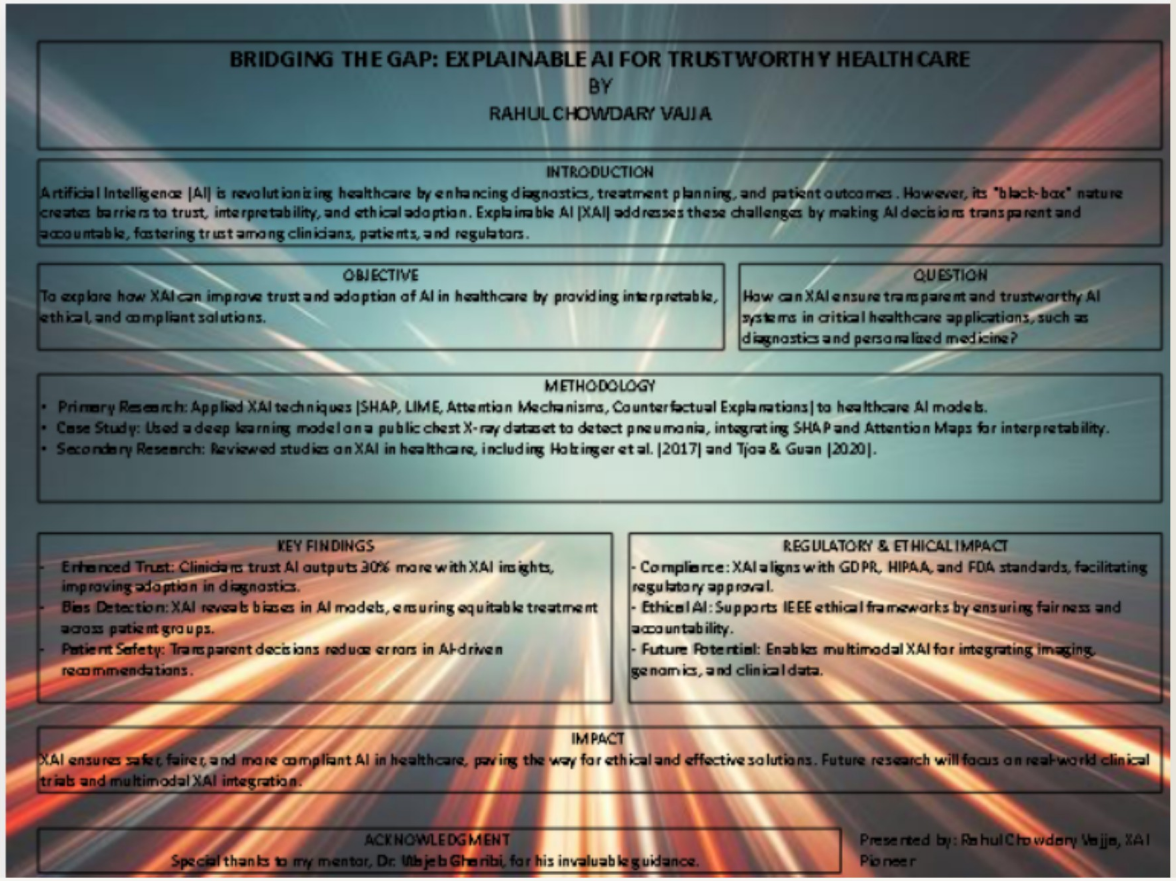
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Now some that did not win

- The project may have been good.
- It is hard to find that out in the minute or two a judge spends at a poster.

a passerby decides	from
whether to walk over	the look of it from 10 feet away
whether to read	the title and the headings
whether to ask	the results they can spot

Hard to read



BRIDGING THE GAP: EXPLAINABLE AI FOR TRUSTWORTHY HEALTHCARE

BY

RAHUL CHOWDARY VAJJA

INTRODUCTION

Artificial Intelligence (AI) is revolutionizing healthcare by enhancing diagnostics, treatment planning, and patient outcomes. However, its "black-box" nature creates barriers to trust, interpretability, and ethical adoption. Explainable AI (XAI) addresses these challenges by making AI decisions transparent and accountable, fostering trust among clinicians, patients, and regulators.

OBJECTIVE

To explore how XAI can improve trust and adoption of AI in healthcare by providing interpretable, ethical, and compliant solutions.

QUESTION

How can XAI ensure transparent and trustworthy AI systems in critical healthcare applications, such as diagnostics and personalized medicine?

METHODOLOGY

- Primary Research: Applied XAI techniques (SHAP, LIME, Attention Mechanisms, Counterfactual Explanations) to healthcare AI models.
- Case Study: Used a deep learning model on a public chest X-ray dataset to detect pneumonia, integrating SHAP and Attention Maps for interpretability.
- Secondary Research: Reviewed studies on XAI in healthcare, including Holzinger et al. (2017) and Tjoa & Guan (2020).

KEY FINDINGS

- Enhanced Trust: Clinicians trust AI outputs 30% more with XAI insights, improving adoption in diagnostics.
- Bias Detection: XAI reveals biases in AI models, ensuring equitable treatment across patient groups.
- Patient Safety: Transparent decisions reduce errors in AI-driven recommendations.

REGULATORY & ETHICAL IMPACT

- Compliance: XAI aligns with GDPR, HIPAA, and FDA standards, facilitating regulatory approval.
- Ethical AI: Supports IEEE ethical frameworks by ensuring fairness and accountability.
- Future Potential: Enables multimodal XAI for integrating imaging, genomics, and clinical data.

IMPACT

XAI ensures safer, fairer, and more compliant AI in healthcare, paving the way for ethical and effective solutions. Future research will focus on real-world clinical trials and multimodal XAI integration.

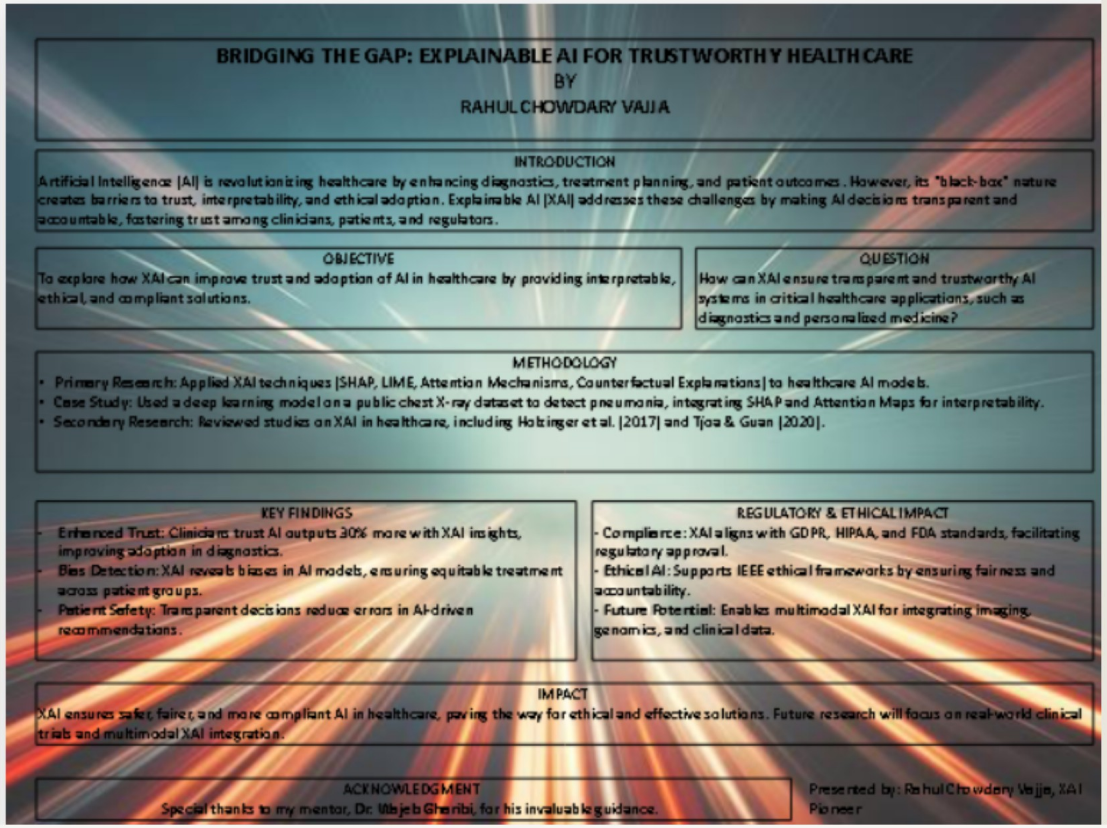
ACKNOWLEDGMENT

Special thanks to my mentor, Dr. Wajeb Gharibi, for his invaluable guidance.

Presented by: Rahul Chowdary Vajja, XAI Pioneer

Hard to read

- Hard to read.
- No visual elements.
- Poor contrast.



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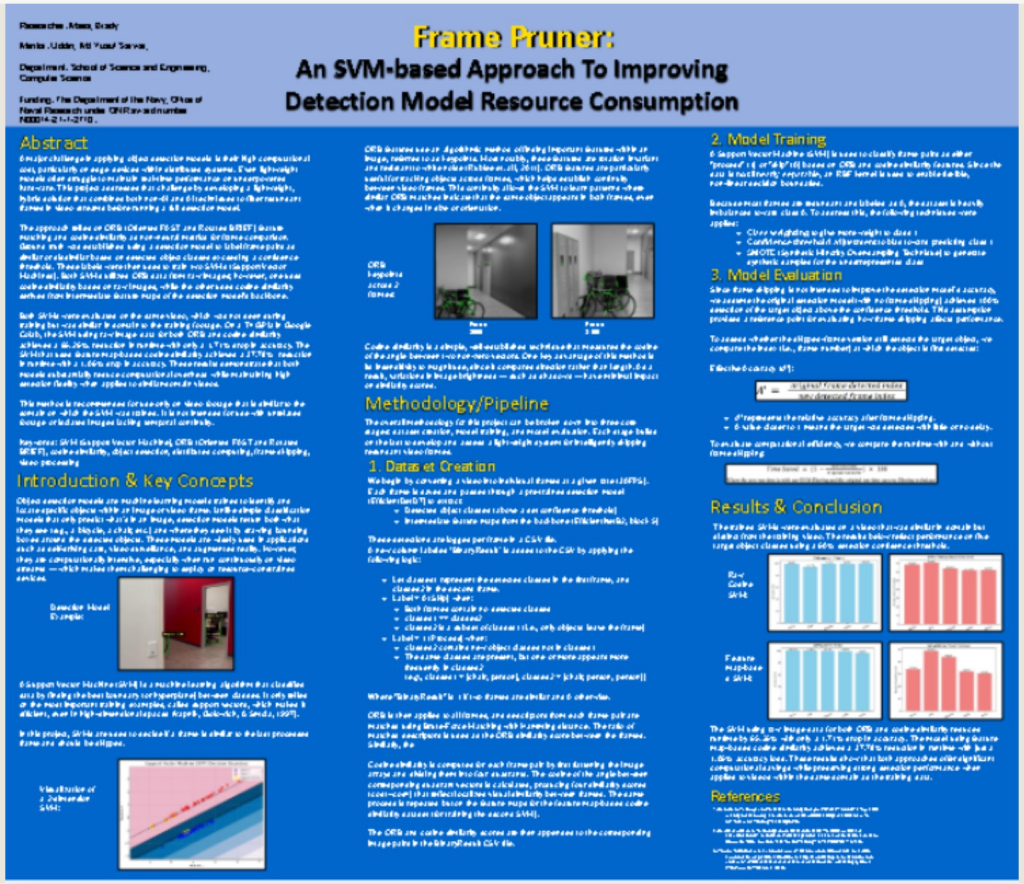
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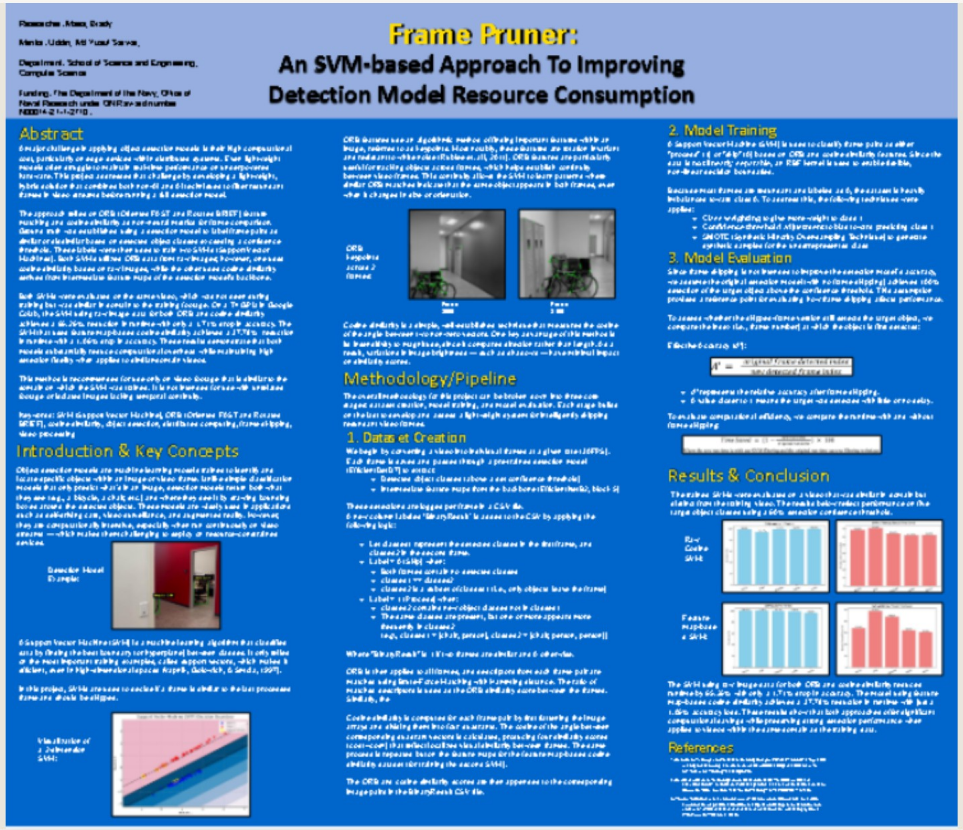
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Nothing separates the columns

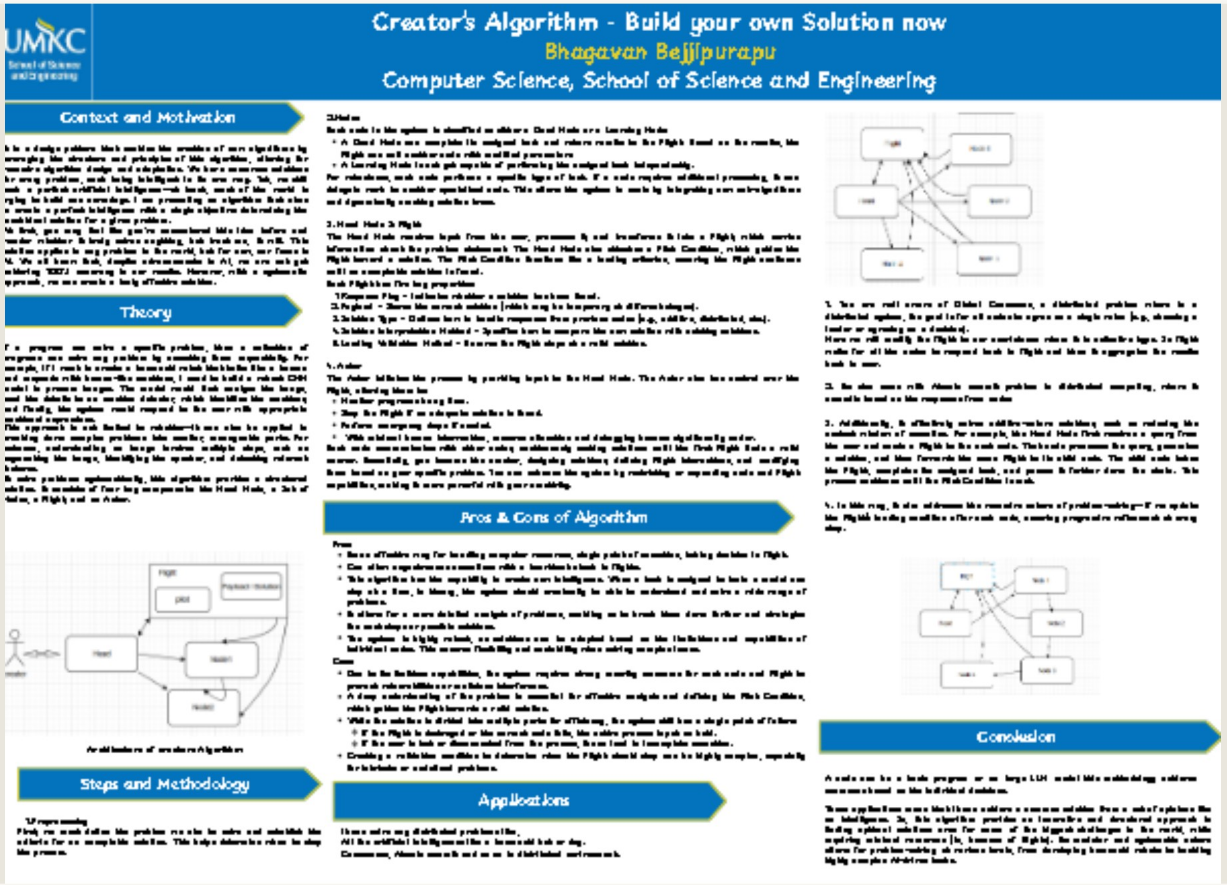


Nothing separates the columns

- Poor color contrast for the text.
- No boxes or header bars to break apart the columns.
- Cutting some text would have made room for larger images, or more of them.

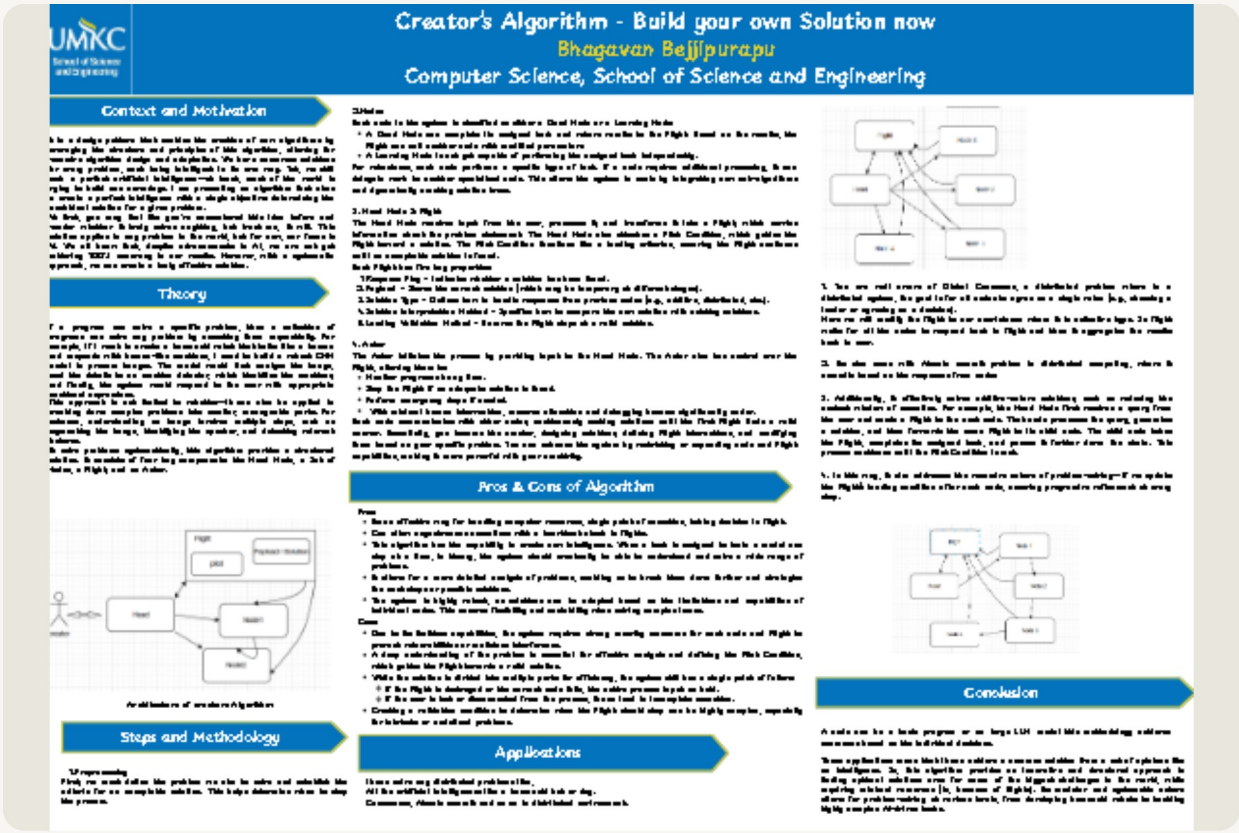


A wall of text

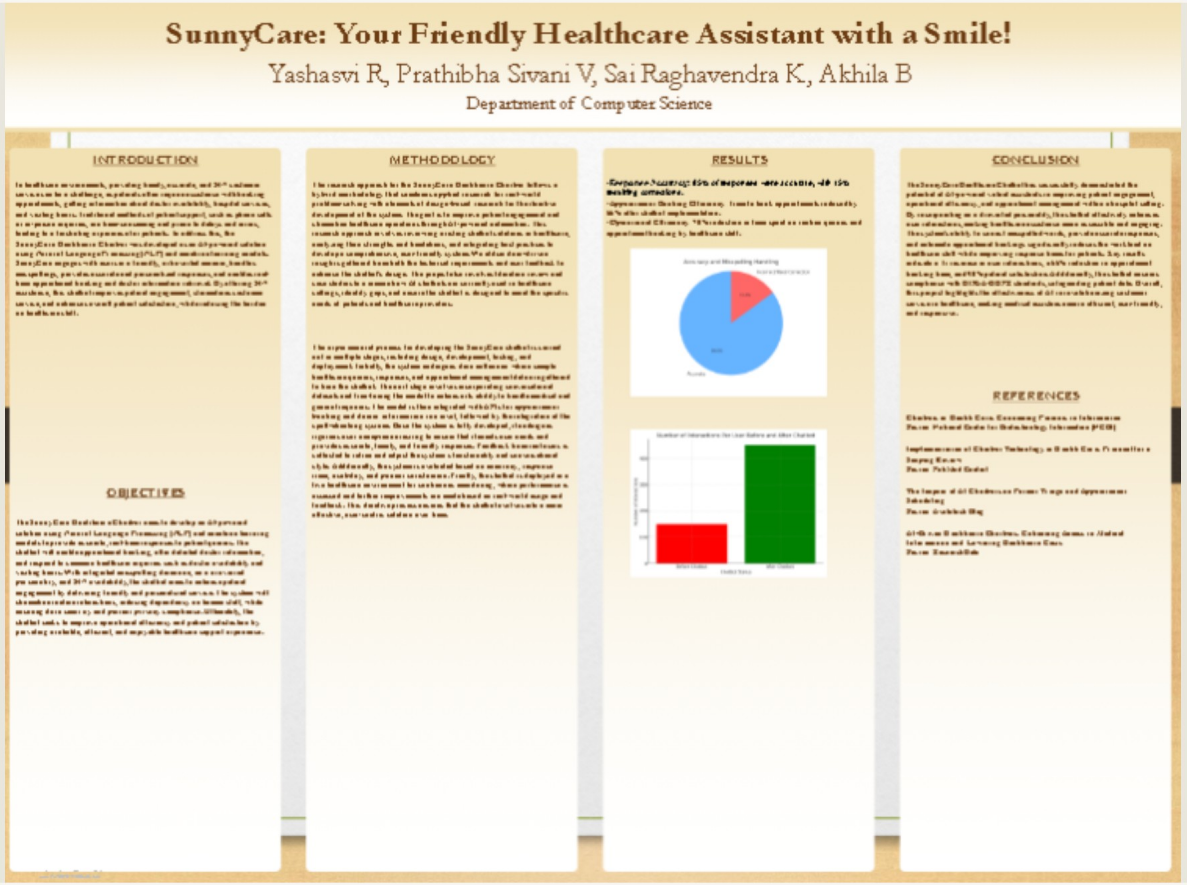


A wall of text

- Wall of text. No color.
- Nothing grabs the eye.
- Very little negative space.
- It feels intimidating to approach. Where do I start? This is going to take focus to understand.

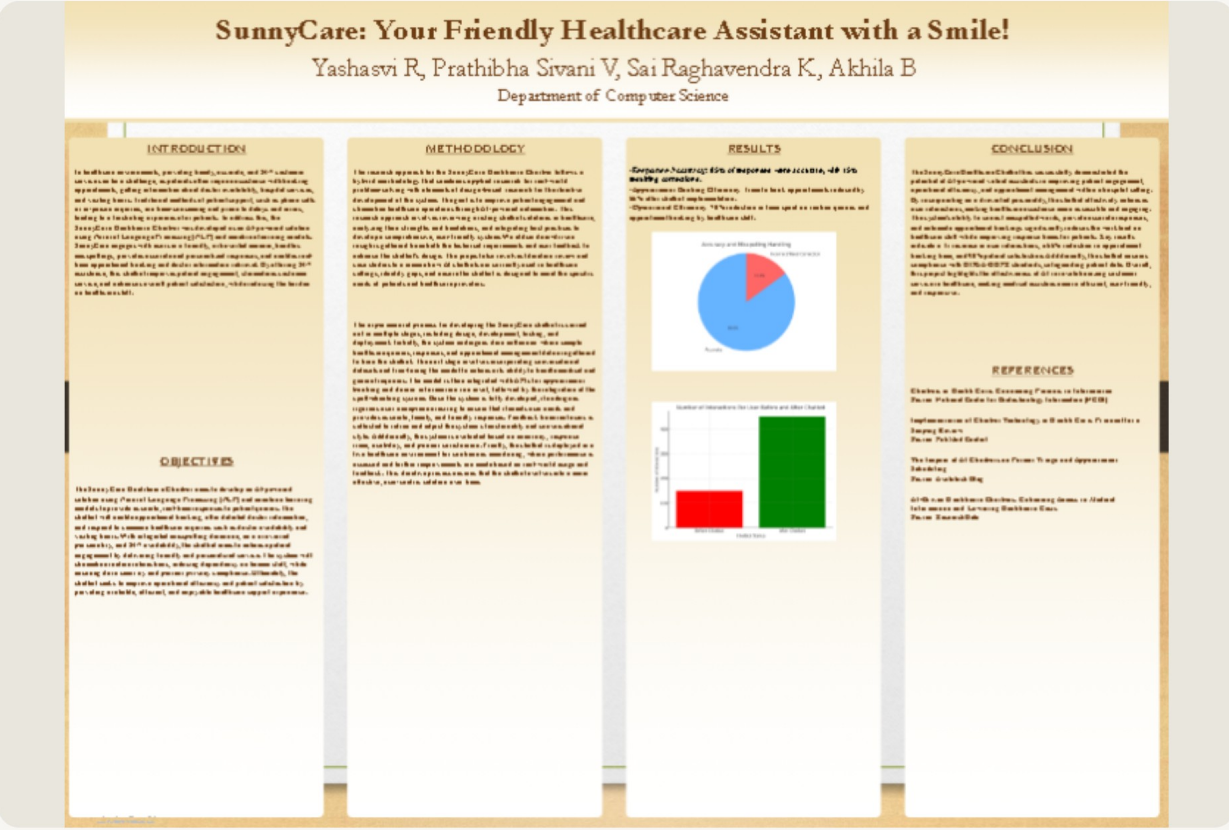


Unused space

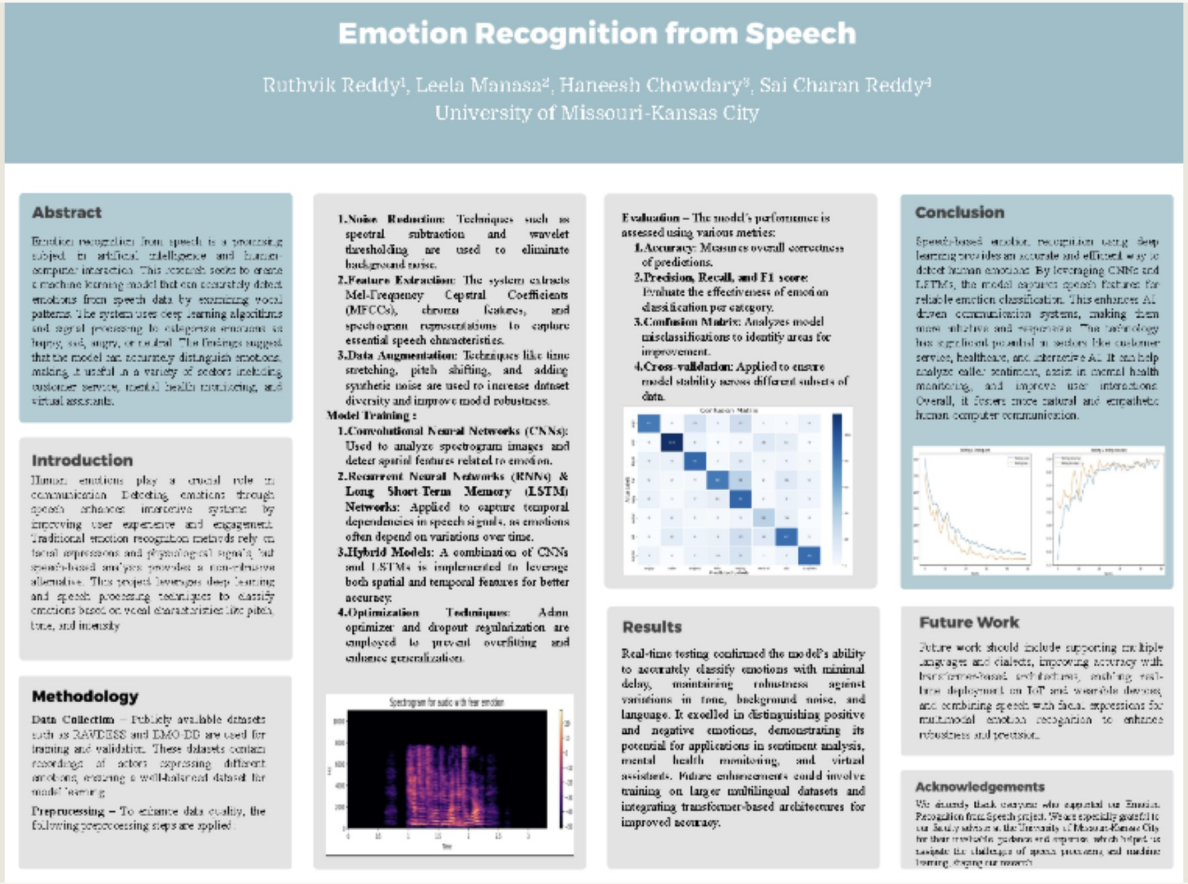


Unused space

- Poor color contrast.
- They did not use the available space. It reads as low effort.
- You want negative space, not huge blank areas.
- Two simple visuals.



Too subdued



Emotion Recognition from Speech

Ruthvik Reddy[1], Leela Manasa[2], Haneesh Chowdary[3], Sai Charan Reddy[4]
University of Missouri-Kansas City

Abstract

Emotion recognition from speech is a promising subject in artificial intelligence and human-computer interaction. This research seeks to create a machine learning model that can accurately detect emotions from speech data by examining vocal patterns. The system uses deep learning algorithms and signal processing to categorize emotions as happy, sad, angry, or neutral. The findings suggest that the model can accurately distinguish emotions, making it useful in a variety of sectors including customer service, mental health monitoring, and virtual assistants.

Introduction

Human emotions play a crucial role in communication. Detecting emotions through speech enhances interactive systems by improving user experience and engagement. Traditional emotion recognition methods rely on facial expressions and physiological signals, but speech-based analysis provides a non-intrusive alternative. This project leverages deep learning and speech processing techniques to classify emotions based on vocal characteristics like pitch, tone, and intensity.

Methodology

Data Collection – Publicly available datasets such as RAVDESS and EMO-DB are used for training and validation. These datasets contain recordings of actors expressing different emotions, ensuring a well-balanced dataset for model training.

Preprocessing – To enhance data quality, the following preprocessing steps are applied:

1. **Noise Reduction:** Techniques such as spectral subtraction and wavelet thresholding are used to eliminate background noise.
2. **Feature Extraction:** The system extracts Mel-Frequency Cepstral Coefficients (MFCCs), chroma features, and spectrogram representations to capture essential speech characteristics.
3. **Data Augmentation:** Techniques like time stretching, pitch shifting, and adding synthetic noise are used to increase dataset diversity and improve model robustness.

Model Training :

1. **Convolutional Neural Networks (CNNs):** Used to analyze spectrogram images and detect spatial features related to emotion.
2. **Recurrent Neural Networks (RNNs) & Long Short Term Memory (LSTM) Networks:** Applied to capture temporal dependencies in speech signals, as emotions often depend on variations over time.
3. **Hybrid Models:** A combination of CNNs and LSTMs is implemented to leverage both spatial and temporal features for better accuracy.
4. **Optimization Techniques:** Adam optimizer and dropout regularization are employed to prevent overfitting and enhance generalization.

Evaluation – The model's performance is assessed using various metrics:

1. **Accuracy:** Measures overall correctness of predictions.
2. **Precision, Recall, and F1 score:** Evaluate the effectiveness of emotion classification per category.
3. **Confusion Matrix:** Analyzes model misclassifications to identify areas for improvement.
4. **Cross-validation:** Applied to ensure model stability across different subsets of data.

Results

Real-time testing confirmed the model's ability to accurately classify emotions with minimal delay, maintaining robustness against variations in tone, background noise, and language. It excelled in distinguishing positive and negative emotions, demonstrating its potential for applications in sentiment analysis, mental health monitoring, and virtual assistants. Future enhancements could involve training on larger multilingual datasets and integrating transformer-based architectures for improved accuracy.

Conclusion

Speech-based emotion recognition using deep learning provides an accurate and efficient way to detect human emotions. By leveraging CNNs and LSTMs, the model captures speech features for reliable emotion classification. This enhances AI-driven communication systems, making them more intuitive and responsive. The technology has significant potential in sectors like customer service, healthcare, and interactive AI. It can help analyze caller sentiment, assist in mental health monitoring, and improve user interactions. Overall, it fosters more natural and empathetic human-computer communication.

Future Work

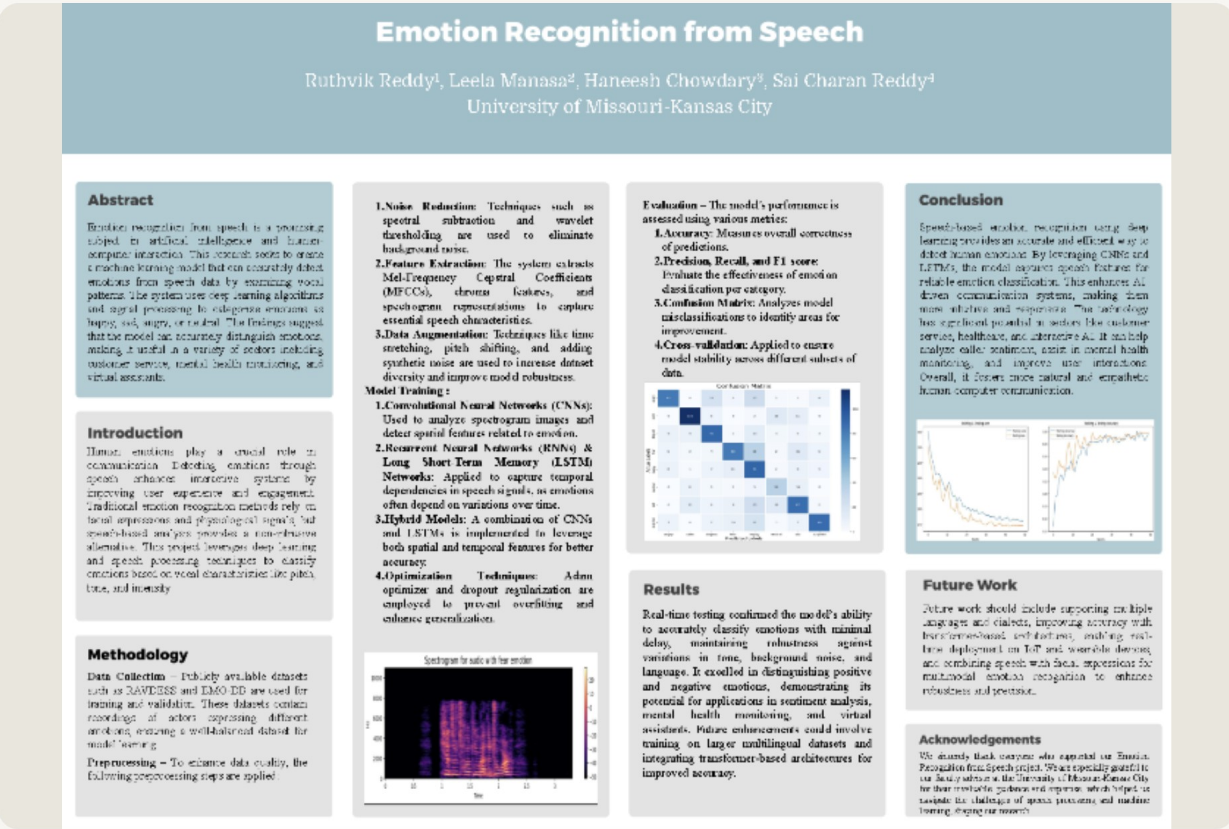
Future work should include supporting multiple languages and dialects, improving accuracy with [illegible] and [illegible], enabling real-time deployment on IoT and wearable devices, and combining speech with facial expressions for multimodal emotion recognition to enhance robustness and precision.

Acknowledgements

We sincerely thank everyone who supported our Emotion Recognition from Speech project. We are especially grateful to our faculty advisor at the University of Missouri-Kansas City for their invaluable guidance and expertise, which helped us navigate the challenges of speech processing and machine learning, shaping our research.

Too subdued

- This one is pretty good. It is just so subdued.
- Nothing is interesting from 10 feet away.
- It could use more visual elements.



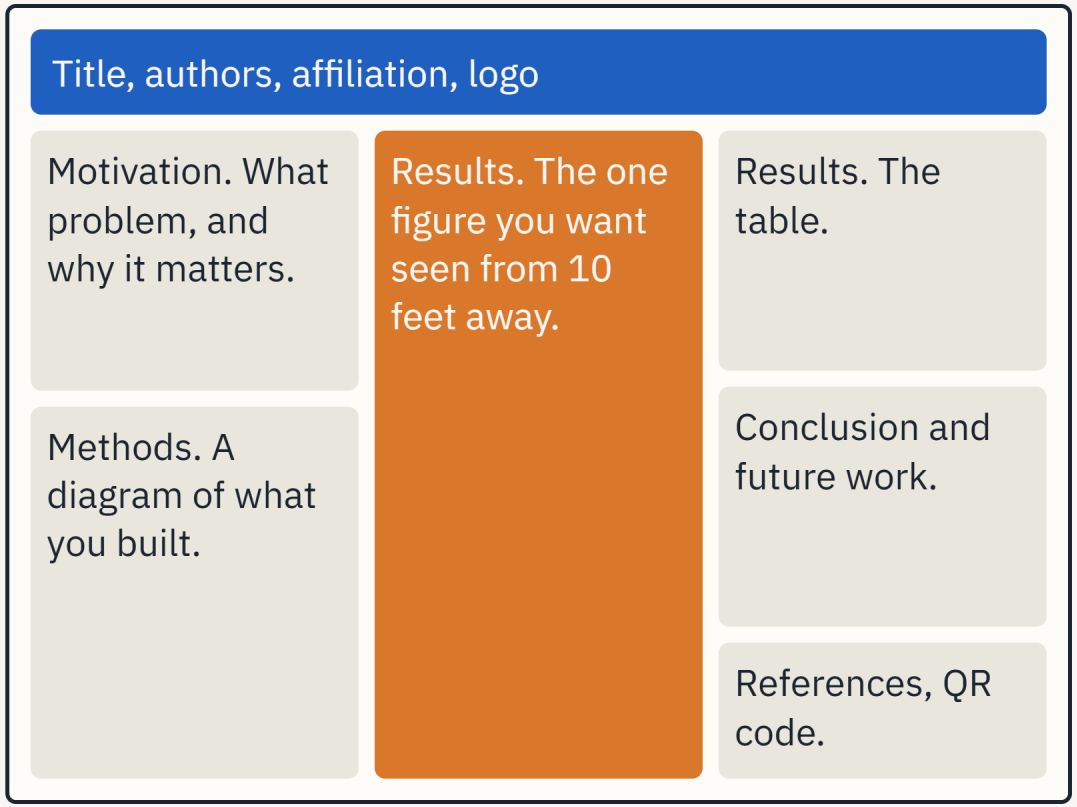
I can almost predict the winners without reading the text

- Every winner passed all five.
- Every poster that did not win failed at least one.
- Check your own poster from across the room before you print it.

#	from 10 feet away
1	more than 5 visual elements
2	legible text with high contrast
3	clear headings and sections
4	enough negative space between elements
5	something interesting

Organize it like a paper

- Same order as a paper: motivation, methods, results, conclusion.
- Readers go left to right, top to bottom. The biggest figure goes in the middle.
- Every box gets a heading. Every heading is readable from the door.



48 by 36 inches, landscape, three columns

Build it in Google Slides

- Google Slides is enough. PowerPoint and Canva work the same way.
- Set the page size first. Resizing later breaks every box.
- It prints at 48 by 36 inches. 24-point body text is readable at arm’s length; the title should read from the door.

step	where
new deck	slides.google.com, blank presentation
page size	File → Page setup → Custom → 48 × 36 inches
layout	Insert → Shape for boxes and header bars, text boxes inside
text size	title 80 pt or more, headings 40 pt or more, body 24 pt or more
figures	Insert → Image at full resolution, not a screenshot of a plot
export	File → Download → PDF, and send the PDF to the printer
check	zoom to 100 percent, then print one page at letter size and read it

Graphics: your own, not generated

- A flowchart you drew looks fine. Your result plots look fine.
- An AI-generated drawing is spottable from across the room, usually has errors, and feels unprofessional.
- Judges notice. It reads as filler where your work should be.

use	avoid
plots of your own results, exported at full resolution	a generated “chart” with made-up numbers
a flowchart or architecture diagram you drew in Slides	a generated diagram with garbled labels
screenshots of your tool, your data, your circuit	a generated illustration of “a scientist” or “a quantum computer”
a photo of real hardware or a real team	stock-style generated art as decoration

The elevator pitch: four beats in under a minute

- You have less than a minute to convince someone your project is cool.
- Motivate the problem. What are you solving, and why does anyone care?
- Say what you did that is interesting, and why.
- Highlight a few findings. Then stop and let them ask.



A rough split. The last beat is the point: the details are on the poster behind you, and a question means the pitch worked.

What not to do

- Do not regurgitate facts, results, or mathematics from the poster.
- Do not give a presentation on the methods.
- You don’t have to be Jim Carrey. You do have to make eye contact and show a little excitement.

instead of	do this
reading the results table aloud	point at it and say what it means in one sentence
walking through the pipeline step by step	one sentence on what you built
reciting every number	the one or two numbers that surprised you
a monotone script with your eyes on the floor	look at them and sound like you like your project
filling the whole minute	stop early and leave room for a question

Practice the beats, not the words

- Memorize the basic points you want to make, in order. Not every word.
- Practice the pitch 5 to 10 times before the event.
- Time it. Under a minute.

a script

Hello, my name is, and our project is titled, and in this project we investigated, using a dataset of, and the methodology we applied was, which consisted of three stages, the first of which...

Forget one word and the whole thing stalls.

beats

1. The problem, one line.
2. What we built, one line.
3. The result that surprised us.
4. Stop.

Say it differently every time. The order is what you keep.